

Student Name: Bak Ji Yong

Student Number: 2023314052

1. External Schema Elements

Name	mods:titleInfo		
Label	Title Information		
URI	http://www.loc.gov/mods/v3#titleInfo		
Source	Mods		
Definition	A structured container for the title of the Japanese anime film. Given the fact that one film can have multiple names across the country because of cultural or commercial reason of each country's, it supports multiple instances to represent the original Japanese title, Romanized transliteration, translated English title, and Korean title. By using the "type" and 'xml:lang' attributes, characteristics of each title are clearly distinguished, enabling multi-language cataloguing within a single repeatable element. To represent transliterated title, use script and transliteration at once, and use authorized transliteration method like Hepburn or Kunrei.		
Data value	Type	xs:string	
	Cardinality	Mandatory / Repeatable	
Subelement (sequence)	Title	Definition	Original title of film
		Type	xs:string
		Cardinality	Mandatory / Unrepeatable
	subTitle	Definition	Subtitle of film
		Type	xs:string
		Cardinality	Optional / Unrepeatable
	partNumber	Definition	Part number of film
		Type	xs:integer
		Cardinality	Optional / Unrepeatable
	partName	Definition	Part name of film
		Type	xs:string
		Cardinality	Optional / Unrepeatable
Attribute	type	Definition	Distinguishes the kind of title

		Type	Enumeration: abbreviated, translated, alternative, or uniform
		Cardinality	Optional / Unrepeatable
	xml:lang	Definition	Distinguishes the language of the title
		Type	External standard (ISO 639 language code)
		Cardinality	Mandatory / Unrepeatable
	script	Definition	Transliterated language (Latin fixed)
		Type	xs:string
		Cardinality	optional / Unrepeatable
	transliteration	Definition	The method of transliteration used (e.g. Hepburn)
		Type	xs:string
		Cardinality	optional / Unrepeatable

Name	schema:director		
Label	Director		
URI	https://schema.org/director		
Source	Schema.org		
Definition	It identifies the director of the anime film. As auteur-driven works are a defining characteristic of the medium (e.g. Hayao Miyazaki, Satoshi Kon, Mamoru Hosoda), it works as major access point in anime film discovery. Although there are many 'directors' in each sector of the movie like painting, sound, story and screenplay, 'director' in this part means only the "General Director". Using Schema.org's established "director" property, maximum interoperability with web-based film databases and search engines could be secured.		
Data value	Type	schema:Person (xs:string name)	
	Cardinality	Mandatory / Repeatable	
Subelements (sequence)	name		
	nameJa		

Attributes	id	Definition	Id of director (for the interoperability with another schema)
		Type	xs:string
		Cardinality	Optional / Unrepeatable

Name	schema:isBasedOn		
Label	Source Work		
URI	https://schema.org/isBasedOn		
Source	Schema.org		
Definition	It identifies the creative work upon which the anime film is based. Critical for distinguishing original productions from adaptations of manga, light novels, visual novels, or video games. Most Japanese animation films are based on pre-existing works, making this element essential for tracing creative lineage. Using enumeration, unity of the system had been ensured. Furthermore, with structure MD, this element has the potential to become a connecting link with the schemas of each domain(manga, novel and etc) in the future.		
Data value	Type	xs:string (title of source work)	
	Cardinality	Optional / Unrepeatable	
Attributes	sourceType	Definition	Categorizes the type of source material.
		Type	Enumeration: manga, novel, light novel, visual novel, game, original, other
		Cardinality	Optional / Unrepeatable

Name	schema:duration		
Label	Runtime Duration		
URI	https://schema.org/duration		
Source	Schema.org		
Definition	It records the theatrical runtime of the anime film in ISO 8601 duration format. Essential for cataloguing, scheduling, and user-facing display. Based on ISO 8601, because there is no need to use one longer than 'day', basic form is PT (which means the start of hour and minute) plus number (only minutes). The reason for not using 'hour' is to avoid inconvenience in comparisons and to standardize units.		

Data value	Type	xs:duration (ISO 8601, e.g. PT119M)
	Cardinality	Mandatory / Unrepeatable

Name	ebucore:hasProducer		
Label	Production Company		
URI	http://www.ebu.ch/metadata/ontologies/ebucore/ebucore#hasProducer		
Source	EBUCore		
Definition	It identifies the animation studio responsible for creating the film. In the Japanese anime industry, the studio is often as significant as the director in shaping a film's visual identity and the direction of the creation. Repeatable for co-productions.		
Data value	Type	xs:string (company name)	
	Cardinality	Mandatory / Repeatable	
Attributes	CompnyID	Definition	Unique identifier or URI for the production company
		Type	xs:anyURI
		Cardinality	Optional / Unrepeatable
	foundedYear	Definition	The year the studio was established.
		Type	xs:gYear
		Cardinality	Optional / Unrepeatable

Name	ebucore:hasPublisher		
Label	Distributor		
URI	http://www.ebu.ch/metadata/ontologies/ebucore/ebucore#hasPublisher		
Source	EBUCore		
Definition	It records the organization responsible for the screening, home video, or streaming distribution. Helpful for understanding business structure of film industry. Repeatable to capture domestic (JP) and international distributors separately.		
Data value	Type	xs:string (distributor name)	
	Cardinality	Mandatory / Repeatable	

Attributes	region	Definition	Geographic place of the market distributor serves
		Type	xs:string (ISO 3166-1)
		Cardinality	Optional / Unrepeatable

Name	mpeg7:Classification		
Label	Classification		
URI	urn:mpeg:mpeg7:schema:2004#Classification		
Source	MPEG-7		
Definition	Structured, multi-level classification. Supports assigning genres, language and parental guidance simultaneously, enabling hierarchical and parallel classification. In genre, enabling multiple selections, it is possible to express multi-layered concept of the film. And, by using the external standard (EBU ContentGenreCS), it can establish a clear genre concept and encourage appropriate usage. In language with xs:language, though most of film in domain has Japanese as primary language, in the case of English film made in Japanese production company and multiple primary language this element could be used. In ParentalGuide, it use official external standard of content rating rule of each country.		
	Type	Complex	
	Cardinality	Mandatory / Unrepeatable	
Subelements (sequence)	Genre	Definition	Specific genre pattern or characteristics of the film. Use a external vocabulary suited for the anime domain.
		Type	xs:string (EBUContentGenreCS)
		Cardinality	Mandatory / Repeatable
	Language	Definition	Records the primary spoken language of the film's original audio track. For this schema's scope, predominantly "ja". Repeatable for films with multiple primary language tracks.
		Type	xs:language (BCP 47 / ISO 639)
		Cardinality	Mandatory / Unrepeatable
	ParentalGuidance	Definition	Official content rating for the film. Each country has its own content rating rule. It could be the form of name of rate or the number itself. Described separately below.
		Cardinality	Optional / Repeatable

Name	mpeg7:ParentalGuidance		
Label	Parental Guidance		
URI	urn:mpeg:mpeg7:schema:2004#ParentalGuidance		
Source	MPEG-7		
Definition	Official content rating for the film. Each country has its own content rating rule. It could be the form of name of rate or the number itself.		
Data value	Type	Complex	
	Cardinality	Mandatory / Repeatable	
Subelements (choice)	ParentalRating	Definition	Parental guidance rate in text format
		Type	xs:string
		Cardinality	Mandatory / Unrepeatable
	MinimumAge	Definition	The minimum age to watch the film
		Type	xs:nonNegativeInteger
		Cardinality	Mandatory / Unrepeatable
Subelements	Region	Definition	Countries where the content rating rule is used
		Type	xs:string (ISO 3166-1)
		Cardinality	Optional / Unrepeatable

Name	mpeg7:KeywordAnnotation		
Label	Keyword Annotation		
URI	urn:mpeg:mpeg7:schema:2004#KeywordAnnotation		
Source	MPEG-7		
Definition	Stores thematic keywords describing the film's content at a granular level — specific motifs, plot devices, and thematic elements that support faceted search and recommendation. For more information and usability same information could be explained in multiple language.		
Data value	Type	Complex	
	Cardinality	Mandatory / Repeatable	
Subelement	Keyword	Definition	Each word in the keywordannotation which represents film's content overall. In Original Mpeg-7, "Keyword" has relevance (with

Attributes			content) and confidence (about the keyword). But it is practically impossible and inefficiency to use and implement in XSD, so in this XSD those two are not adopted
		Type	xs:string
		Cardinality	Mandatory / Repeatable
	xml:lang	Definition	Language of the keyword annotation. Enables separate keyword sets per language.
		Type	xs:language (BCP 47)
		Cardinality	Mandatory / Unrepeatable

Name	dc:date	
Label	Release Date	
URI	http://purl.org/dc/elements/1.1/date	
Source	Dublin Core	
Definition	Records the original Japanese theatrical release date in W3CDTF / ISO 8601 format. Dublin Core defines dc:date as "a point or period of time associated with an event in the lifecycle of the resource." It works as authoritative date for cataloguing and comparing purposes. It accepts either a full date (xs:date) or year only (xs:gYear), in the consideration of convenience	
Data value	Type	Union of xs:date and xs:gYear (ISO 8601, e.g. 2009-08-01 or 2009)
	Cardinality	Mandatory / Unrepeatable

Name	dc:description	
Label	Synopsis / Description	
URI	http://purl.org/dc/elements/1.1/description	
Source	Dublin Core	
Definition	A concise, engaging narrative summary of the film's plot. Dublin Core defines it as "an account of the resource". In domain, this element naturally serves the synopsis role. Written in a style that attracts viewer interest for catalogue browsing and user-facing interfaces like brief introduction of film in OTT sites and movie posters. And surely, it serves multiple language explanation.	
Data value	Type	xs:string (free text)

	Cardinality	Mandatory / Repeatable	
Attributes	xml:lang	Definition	Language of the description text. Enables multilingual synopses (e.g. ja, en, ko).
		Type	xs:language (BCP 47 / ISO 639)
		Cardinality	Mandatory / Unrepeatable

2. Homemade (AF-abb of animation film) Elements

Name	af:voiceCast		
Label	Voice Cast		
URI	http://example.org/anime-film#voiceCast		
Source	Homemade (AF)		
Definition	A container element listing voice actors and their corresponding character roles. Unlike live-action film metadata, anime requires explicit pairing of voice performers and animated characters. Which is one of the great joys of animation film. Each child <voiceActor> entry includes the actor's name and the character they portray, reflecting the central role of voice acting in anime production and fandom.		
Data value	Type	Complex	
	Cardinality	Mandatory / Unrepeatable	
Subelement	voiceActor	Definition	Granular explanation of each voice cast which includes actor's name, character name and their central role. For more understandability and ease, it supports both English and Japanese version at once. Described separately below.
		Type	Complex
		Cardinality	Mandatory / Repeatable

Name	af:voiceActor		
Label	Voice Actor		
URI	http://example.org/anime-film#voiceActor		
Source	Homemade (AF)		
Definition	Granular explanation of each voice cast which includes actor's name, character name and their central role. For more understandability and ease, it supports both English and Japanese version at once.		

Data value	Type	Complex	
	Cardinality	Mandatory / Repeatable	
Subelements (sequence)	actorName	Definition	The voice actor's name in English (romanized). Primary display name for international audiences.
		Type	xs:string
		Cardinality	Mandatory / Unrepeatable
	actorNameJa	Definition	The voice actor's name in Japanese script (kanji/kana). Enables native-language display and accurate identification.
		Type	xs:string
		Cardinality	Optional / Unrepeatable
	characterName	Definition	The name of the animated character voiced by this actor, in English. For the case of one voice actor playing multiple characters, it allows multiple appearances.
		Type	xs:string
		Cardinality	Mandatory / Repeatable
	characterNameJa	Definition	The animated character's name in Japanese script. For the case of one voice actor playing multiple characters, it allows multiple appearances.
		Type	xs:string
		Cardinality	Optional / Repeatable
	role	Definition	Central role of voice casting. In the case of one actor who plays multiple characters, it could appear several times.
		Type	Enumeration: lead, supporting, cameo
		Cardinality	Optional / Repeatable
	personId	Definition	A unique identifier string for the voice actor, enabling cross-referencing across records.
		Type	xs:string
		Cardinality	Optional / Unrepeatable
Attributes			

Name	af:musicComposer
------	------------------

Label	Music Composer		
URI	http://example.org/anime-film#musicComposer		
Source	Homemade (AF)		
Definition	It identifies the composer of the film's original musical score. Music is an integral part of the anime film experience, with composers like Joe Hisaishi (with Studio Ghibli) and Radwimps (with Makoto Shinkai) being closely associated with the artistic identity of the films they score (sometime music itself represents the whole story and emotion of films). And "Music Composer" means "Director" of music parts of the film, in the consideration of efficiency. This element captures the person responsible for the film's soundtrack composition.		
Data value	Type	Complex	
	Cardinality	Optional / Unrepeatable	
Subelements (sequence)	name	Definition	The composer's name in English (romanized). Primary display name for international cataloguing.
		Type	xs:string
		Cardinality	Mandatory / Unrepeatable
	nameJa	Definition	The composer's name in Japanese script (kanji/kana).
		Type	xs:string
		Cardinality	Optional / Unrepeatable
Attributes	personId	Definition	A unique identifier string for the composer, enabling cross-referencing across records.
		Type	xs:string
		Cardinality	Optional / Unrepeatable

Name	af:award		
Label	Award		
URI	http://example.org/anime-film#award		
Source	Homemade (AF)		
Definition	It records an award or recognition film received. Structured with year and category attributes to go beyond a simple text string, it enables chronological sorting and category-based filtering. Repeatable for films that receive multiple awards across different festivals and ceremonies.		
Data value	Type	xs:string (award-granting name, e.g. "Japan Academy Prize")	

	Cardinality	Optional / Repeatable	
Attributes	year	Definition	The year the award was granted.
		Type	xs:gYear
		Cardinality	Mandatory / Unrepeatable
	category	Definition	The specific award category (e.g. "Animation of the Year")
		Type	xs:string
		Cardinality	Optional / Unrepeatable

Name	af:animationStyle		
Label	Animation Style		
URI	http://example.org/anime-film#animationStyle		
Source	Homemade (AF)		
Definition	It describes the animation technique and visual style of the film. This is the most domain-specific element in the schema. Because live-action films have cinematography, in contrast, only animation has the concept of drawn visual style, character design proportions, color palette choices, and studio-specific linework aesthetics. Even some films are famous with its tremendous and beautiful animation style (e.g. Akira or Vampire Hunter D) or just one character design (No-face-Kaonashi-from Spirited Away). Each animation studio, animator director develops a distinctive visual identity and technique that is as recognizable as a director's signature, making this element essential for cataloguing and comparing works within the Japanese anime tradition.		
Data value	Type	xs:string (free-text visual style description)	
	Cardinality	Optional / Unrepeatable	
Attributes	technique	Definition	Primary animation production technique used in the film.
		Type	Enumeration: hand-drawn, CGI, hybrid, rotoscope, stop-motion, cel
		Cardinality	Optional / Unrepeatable